

AI Voice Recorder Device Development Requirements (RFP / PRD)

1. Project Overview

Objective

We are developing a portable AI voice recorder that records meetings and conversations, securely uploads audio to the cloud, and provides AI-powered transcription, speaker identification, summaries, and chat through a mobile application.

2. Scope of Work

We require support for:

- Electronic hardware design
 - PCB design
 - Component selection
 - Firmware support
 - Mechanical enclosure design
 - Prototype manufacturing
 - DFM (Design for Manufacturing)
 - EVT/DVT/PVT validation
 - Production manufacturing
 - Testing
 - Certifications (BIS/FCC/CE as applicable)
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3. Product Overview

Device Type

Portable AI Voice Recorder

Primary Use Cases

- Meeting recording
- Voice notes
- Interview recording
- Lecture recording
- Business discussions

4. Expected Product Features

Recording

- Start recording (Physical Button & Mobile App via BLE)
 - Stop recording (Physical Button & Mobile App via BLE)
 - Pause / Resume recording (Physical Button & Mobile App via BLE)
 - Local storage
 - Continuous recording
 - Long-duration recording
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Audio

- High-quality MEMS microphones
 - Noise reduction
 - Echo cancellation (optional)
 - Voice clarity optimization
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Storage

- Local storage
 - Secure file management
 - Automatic file organization
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Connectivity

BLE

- Device discovery
- Pairing
- Wi-Fi provisioning
- Device configuration
- Device diagnostics

Wi-Fi

- Audio upload

- OTA updates
 - Device synchronization
 - Secure cloud communication
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Power

- Rechargeable battery
 - USB-C charging
 - Battery monitoring
 - Deep sleep
 - Low power operation
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Security

- Secure storage
 - Device authentication
 - Encrypted communication
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5. Mobile Application Integration

The device must communicate with our mobile application using BLE.

Required BLE operations:

- Device pairing
- Start/Stop/Pause/Resume recording
- Real-time recording status
- Recording duration
- User authentication
- Wi-Fi provisioning
- Device settings
- Battery status
- Recording status
- Firmware updates

6. Cloud Integration

The device should communicate with our backend APIs over HTTPS.

Functions include:

- Authentication
- Upload recordings
- Download configuration
- OTA firmware
- Device status reporting

Backend APIs will be provided by us.

7. Device Functional Requirements

Recording

- Start
 - Stop
 - Pause
 - Resume
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Storage

- Save recordings
 - Delete recordings
 - Storage information
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Battery

- Battery percentage
 - Charging detection
 - Low battery alerts
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LEDs

- Recording
 - Charging
 - Uploading
 - Error indication
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Buttons

- Record
 - Power
 - Reset
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OTA

- Firmware updates
 - Rollback support
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8. Connectivity Requirements

BLE

- Device discovery & pairing
 - User authentication
 - Provision Wi-Fi
 - Configure device
 - Read device information
 - Start / Stop / Pause / Resume recording
 - Battery & storage status
 - Firmware updates
 - Diagnostics
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Wi-Fi

- Connect to internet
- Upload recordings

- Sync configuration
 - OTA updates
 - Status reporting
 - Retry failed uploads
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Concurrent Operations

- Recording audio
 - BLE connection
 - Wi-Fi connectivity
 - Saving recordings
 - Uploading recordings
 - Battery monitoring
 - Device status synchronization
 - Background processing & retries
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Performance Requirements

- 8–12 hours continuous recording
 - Stable long-duration recording
 - Fast startup and response
 - Reliable BLE & Wi-Fi connectivity
 - Concurrent task execution
 - Low power consumption
 - Automatic recovery & retry
 - Secure OTA updates
 - High system reliability
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9. Hardware Expectations

We expect recommendations for:

- MCU

- Microphones
 - Battery
 - Charger IC
 - PMIC
 - BLE/Wi-Fi antenna
 - Flash memory
 - Storage
 - USB-C implementation
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10. Mechanical Design

Expected support:

- Industrial design
 - 3D CAD
 - Injection mold design
 - Waterproofing (optional)
 - Drop resistance
 - Thermal design
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11. Prototype Deliverables

We expect:

- Schematics
- PCB Layout
- Gerber files
- BOM
- Firmware source
- Mechanical CAD
- 3D models
- Prototype units
- Test reports

12. Manufacturing Deliverables

- PCB fabrication
- SMT assembly
- Firmware flashing
- Functional testing
- Packaging
- Production testing

13. Certification Support

Expected assistance with:

- BIS
 - FCC
 - CE
 - RoHS
 - Battery safety
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